



**ANGUS
FIRE**



RespondolATF

Next generation, alcohol resistant,
fluorine free foam.



EN
1568



Respondol^{ATF}

Respondol ATF continues to push the boundaries of modern-day fluorine free technology by delivering outstanding performance in fresh and sea water, whilst balancing environmental impact.

Respondol ATF is the only fluorine free foam in the world with UL listings for ethanol, methanol and up to a 15% blend of ethanol and petrol/gasoline. Furthermore Respondol ATF is approved to the IMO, International Maritime Organisation. It doesn't stop there, Respondol ATF has also passed Lastfire with excellent results not just in fresh water but also sea water.

Respondol ATF is a superior quality synthetic fluorine free (FF) foam concentrate, designed for extinguishing and securing all types of flammable liquid fires and Class A incidents. Respondol ATF has been designed specifically for general emergency responders who are faced with a variety of risks in a range of situations.

- Fast knock-down and extinguishment exceeding the requirements of EN1568 pt 3 & 4, UL162, IMO and Lastfire.
- Lower viscosity than other fluorine free foams to ensure easy induction.
- Fluorine free to eliminate environmental persistence.
- Various induction rates for both hydrocarbon and polar solvent risks.



High performance foam with minimal environmental impact

Respondol ATF is a unique, patented formulation with no intentionally added fluorosurfactants, fluoropolymers, PFAS, PFOS, or PFOA. Respondol ATF is 100% biodegradable and especially suited to emergency incidents where containment of firewater run-off cannot be guaranteed as required by legislation and new treaties around the world.



Ideal for high risk situations

Respondol ATF is ideal for use during Class A incidents involving solid materials such as wood, paper or textiles and Class B emergencies involving flammable liquids such as petrol, diesel or oils. It can also be used on polar solvents such as alcohol, ketones, esters and ethers. Respondol ATF brings versatility to first responders when they need it most; during an emergency incident.



Effective and efficient

Respondol ATF provides a vapour suppressing foam blanket on unignited hydrocarbon spills and polar solvents. It produces a vapour sealing blanket of foam that rapidly spreads over the surface of the fuel to efficiently control and extinguish the blaze. It is intended for use at 3% or 6%



Testing and Listing

- EN1568 part 3 and 4 on fresh water/sea water (1A/1A, 1A/1A, 1A/1A).
- UL162 listed on heptane, 15% ethanol/gasoline blend, IPA, ethanol and methanol.
- IMO MSC.1/Circ.1312.
- Lastfire batch certified across all 6 tests with all devices in fresh and sea water.
- TOP-A testing and PFAS analysis all analytes <LOD.



Equipment

Respondol ATF is suitable for use with air aspirating and some non-air aspirating discharge devices such as low expansion branchpipes, standard equipment and other apparatus. It is important to note, as with all non-film forming foams, application rates and application techniques are critical. Please contact Angus Fire for more information.

RespondolATF

For more information on Respondol ATF,
or other Angus Fire products,
please visit: www.angusfire.co.uk
or call +44 (0) 1524 264000



Angus Fire is a global leader in
firefighting technology.

In more than 100 countries Angus Fire supplies fire safety products and services to customers operating in a wide range of industries such as oil and gas, international airports, harbours, ports, military bases, power stations, and fire and rescue services. Angus Fire is a global name with an impressive history of over 220 years in the firefighting industry. It is this rich heritage and associated expertise, which makes Angus Fire the preferred partner with firefighters worldwide.

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Respondol ATF 3-3%

Fluorine Free (FF)
Foam Concentrate

Integrity

*Doing what's right, rather than
what's convenient*

Angus Fire prides itself on the open and honest way in which we conduct our business throughout the world. Our foams are an extension of our ethical beliefs and we pride ourselves in being the responsible foam manufacturer, balancing high performance with minimal environmental impact. Our fluorine free formulations contain no fluorosurfactants, fluoropolymers, organohalogens, PFAS, PFOA and no PFOS in accordance with EU Directive 2006/122/EC and amended Council Directive 76/769/EEC.

Patented Foam Technology

Since Angus Fire became the first foam manufacturer to offer a commercially available fluorine free foam, our chemists have worked hard to always push the boundaries of fluorine free technology. Respondol foam concentrate has been designed as a multi-purpose fluorine free foam for those circumstances when minimising environmental impact is paramount.

A unique patented formulation – repeatable and responsible certification



- High approval rating under EN1568 pt 3 & 4 on all fuels using all waters; 1A/1A – 1A/1A – 2A/2A.
- Lower viscosity than other fluorine free foams to ensure easy induction.
- Fluorine free to minimise environmental persistence.
- 3% induction on both hydrocarbon and polar solvent risks.

Respondol ATF 3-3 is a superior quality 3% synthetic fluorine free (FF) foam concentrate, designed for extinguishing and securing all types of flammable liquid fires and Class A incidents. Respondol ATF 3-3 has been designed specifically for emergency responders who are faced with a variety of risks in a range of situations.

Respondol ATF 3-3 is a patented combination of surfactants and other ingredients to produce a vapour sealing blanket of foam that rapidly spreads over the surface of the fuel to provide rapid control and extinguishment.

- Unique patented formulation only available from Angus Fire.
- Specifically designed for those emergency responders under environmental pressures.
- Approved to EN1568 part 3 and 4 on all fuels and all water types.

Applications

Respondol ATF 3-3 is used in high risk situations where hydrocarbons (such as

oils, gasoline, diesel fuel, and aviation kerosene) are stored, processed, or transported and/or polar solvents (such as alcohols, ketones, esters, and ethers) are stored, processed, or transported.

Respondol ATF 3-3 provides a vapour suppressing foam blanket on unignited hydrocarbon spills.

Respondol ATF 3-3 can also be used as a wetting agent in combating fires in Class A materials such as wood, paper, and tyres.

Approvals and listings

Respondol ATF 3-3 is independently tested and certified to EN1568:2008 part 3 and 4. Performance exceeds the requirements of these tests.

Respondol ATF 3-3 is audited and approved to Underwriters Laboratories UL162 (7th Edition).

Respondol ATF 3-3 also meets the requirements of IMO MSC.1/Circ.1312.

Respondol ATF 3-3 passed Lastfire with excellent results in fresh and sea water.

Equipment

Respondol ATF 3-3 is intended for use at 3% on hydrocarbons and polar solvents. It is readily proportioned using conventional foam proportioning equipment such as portable and fixed (in-line) foam venturi proportioners. Respondol ATF 3-3 should be used with air aspirating discharge devices such as low expansion branchpipes and other devices.

Respondol ATF 3-3%

Fluorine Free (FF) Foam Concentrate

Compatibility

Respondol ATF 3-3 is suitable for use in combination with:

- Soft or hard, fresh, brackish or sea water.
- Expanded protein-based or synthetic foams for application to a fire in sequence or simultaneously.

Environment

Respondol ATF 3-3 is PFOS free in accordance with EU Directive 2006/122/EC and amended Council Directive 76/769/EEC. Respondol ATF 3-3 is 100% biodegradable and is manufactured without any added fluorinated surfactants or fluorinated polymers.

Storage

Respondol ATF 3-3 is exceptionally stable in long-term storage. A shelf-life of at least ten years can be expected if it is stored correctly.

Disposal

For fire water runoff and accidental spillage please refer to Angus Fire's Foam Disposal Guide and MSDS for more information.

Product Quality

Respondol ATF 3-3 production is closely controlled, Angus Fire operates a quality management system which complies with the requirements of BS EN ISO 9001 and BS EN ISO 14001.

Typical Physico-Chemical Properties

Appearance		Light Yellow
Specific gravity @ 20°C (68°F)		1.00 - 1.04
pH @ 20°C (68°F)		7 - 8
Viscosity @ 20°C (68°F)	cP	Non-newtonian
Maximum continuous storage temperature	°C (°F)	49 (120)
Maximum intermittent storage temperature	°C (°F)	60 (140)
Freezing point	°C (°F)	-6 (21.2)
Effect of freeze/thaw		No loss of performance
Lowest use temperature	°C (°F)	1.7 (35)

Typical Foam Properties:

Foam generated using the U.K. Defence Standard DEF42-40 5 lpm branchpipe at 7 Bar pressure.
Foam collected in a 1630 ml N.F.P.A. drainage pan.

Induction rate		3
Expansion ratio		≥ 7:1
25% drainage time	hour/min/sec	≥ 1'00'00"

Typical Packing Specification

	Plastic Square	Plastic Square	Plastic Cylindrical	Plastic Cylindrical	Ecobulk MX
Capacity	25 litres	5 US gallons	200 litres	55 US gallons	1000 litres
Empty weight (kg)	1.2	0.8	9.0	9.0	70
Filled weight (kg)	27	20	215	223	1100
Dimensions (mm)	448 x 286 x 286	402 x 293 x 240	580 D x 922 H	580 D x 922 H	1200 L x 1000 W x 1160 H
Part number	FN0521F0P	FN0521G0P	FN0521J0P	FN0521W0P	FN0521L8



**EN1568:2008
Parts 3 & 4**

EMERGENCY FOAM SERVICE Call +44 (0) 15242 61166 – 24 hours a day, every day

INTERNATIONAL SALES

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Angus Fire operates a continuous programme of product development. The right is therefore reserved to modify any specification without prior notice and Angus Fire should be contacted to ensure that the current issues of all technical data sheets are used.

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Respondol ATF 3-6%

Fluorine Free (FF)
Foam Concentrate

Integrity

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what's convenient*

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A unique patented formulation – repeatable and responsible certification.



- Highest approval rating under EN1568 pt 3 & 4 on all fuels using all waters; 1A/1A – 1A/1A – 1A/1A.
- Lower viscosity than other fluorine free foams to ensure easy induction.
- Fluorine free to minimise environmental persistence.
- 3% induction on hydrocarbon and 6% for polar solvent risks.

Respondol ATF 3-6 is a superior quality synthetic fluorine free (FF) foam concentrate, designed for extinguishing and securing all types of flammable liquid fires and Class A incidents. Respondol ATF 3-6 has been designed specifically for emergency responders who are faced with a variety of risks in a range of situations.

Respondol ATF 3-6 is a patented combination of surfactants and other ingredients to produce a vapour sealing blanket of foam that rapidly spreads over the surface of the fuel to provide rapid control and extinguishment.

- Unique patented formulation only available from Angus Fire.
- Specifically designed for those emergency responders under environmental pressures.
- Approved to EN1568 part 3 and 4 on all fuels and all water types.

Applications

Respondol ATF 3-6 is used in high risk situations where hydrocarbons (such as oils, gasoline, diesel fuel, and aviation kerosene) are stored, processed, or transported and/or polar solvents (such as alcohols, ketones, esters, and ethers) are stored, processed, or transported.

Respondol ATF 3-6 provides a vapour suppressing foam blanket on unignited hydrocarbon spills.

Respondol ATF 3-6 can also be used as a wetting agent in combating fires in Class A materials such as wood, paper, and tyres.

Approvals and listings

Respondol ATF 3-6 is independently tested and certified to EN1568:2008 part 3 and 4.

Performance exceeds the requirements of these tests. Please ask your Angus Fire representative for classifications against this standard.

Equipment

Respondol ATF 3-6 is intended for use at 3% (3 parts concentrate to 97 parts of water) on hydrocarbons and 6% (6 parts concentrate to 94 parts of water) on polar solvents. Respondol ATF 3-6 is readily proportioned using conventional foam proportioning equipment such as portable and fixed (in-line) foam venturi proportioners, handline nozzles/branchpipes with pick-up tubes, balanced pressure variable flow proportioners, balanced

Respondol ATF 3-6%

Fluorine Free (FF) Foam Concentrate

pressure bladder tank proportioners,
and around-the-pump proportioners.

Compatibility

Respondol ATF 3-6 is suitable for use in combination with:

- Soft or hard, fresh, brackish or sea water.
- Expanded protein-based or synthetic foams for application to a fire in sequence or simultaneously.

Environment

Respondol ATF 3-6 is PFOS free in accordance with EU Directive 2006/122/EC and amended Council Directive 76/769/EEC. Respondol ATF 3-6 is 100% biodegradable and is manufactured without any added fluorinated surfactants or fluorinated polymers.

Storage

Respondol ATF 3-6 is exceptionally stable in long-term storage. A shelf-life of at least ten years can be expected if it is stored correctly.

Disposal

For fire water runoff and accidental spillage please refer to Angus Fire's Foam Disposal Guide and MSDS for more information.

Product Quality

Respondol ATF 3-6 production is closely controlled, Angus Fire operates a quality management system which complies with the requirements of BS EN ISO 9001 and BS EN ISO 14001.

Typical Physico-Chemical Properties

Appearance		Light Yellow
Specific gravity @ 20°C (68°F)		1.00 - 1.04
pH @ 20°C (68°F)		7 - 8
Viscosity @ 20°C (68°F)	cP	Non-newtonian
Maximum continuous storage temperature	°C (°F)	49 (120)
Maximum intermittent storage temperature	°C (°F)	60 (140)
Freezing point	°C (°F)	-6 (21.2)
Effect of freeze/thaw		No loss of performance
Lowest use temperature	°C (°F)	-5 (23)

Typical Foam Properties:

Foam generated using the U.K. Defence Standard DEF42-40 5 lpm branchpipe at 7 Bar pressure.
Foam collected in a 1630 ml N.F.P.A. drainage pan.

Induction rate	3	6
Expansion ratio	≥ 7:1	≥ 7:1
25% drainage time	hour/min/sec	≥ 1'00'00" ≥ 3'00'00"

Typical Packing Specification

	Plastic Square	Plastic Square	Plastic Cylindrical	Plastic Cylindrical	Ecobulk MX
Capacity	25 litres	5 US gallons	200 litres	55 US gallons	1000 litres
Empty weight (kg)	1.2	0.8	9.0	9.0	70
Filled weight (kg)	27	20	215	223	1100
Dimensions (mm)	448 x 286 x 286	402 x 293 x 240	580 D x 922 H	580 D x 922 H	1200 L x 1000 W x 1160 H
Part number	FN0522F0P	FN0522G0P	FN0522J0P	FN0521W0P	FN0522L8



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